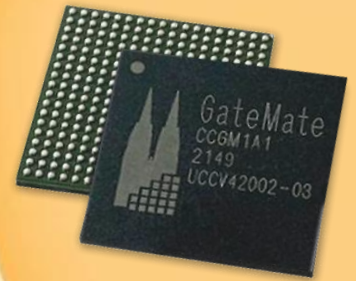


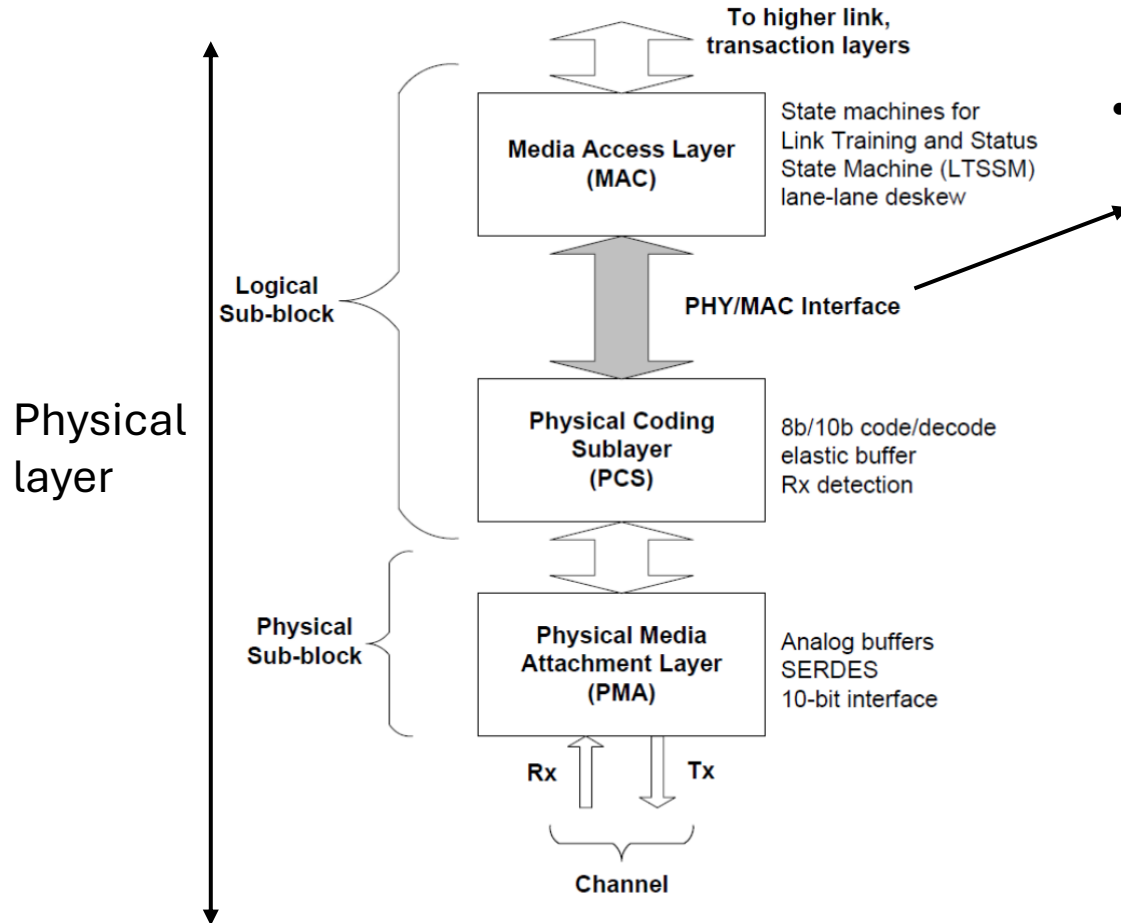
GateMate™ FPGA

MAC layer

Vinh Hai Luong, B.Sc.
Research&Development
Cologne Chip AG

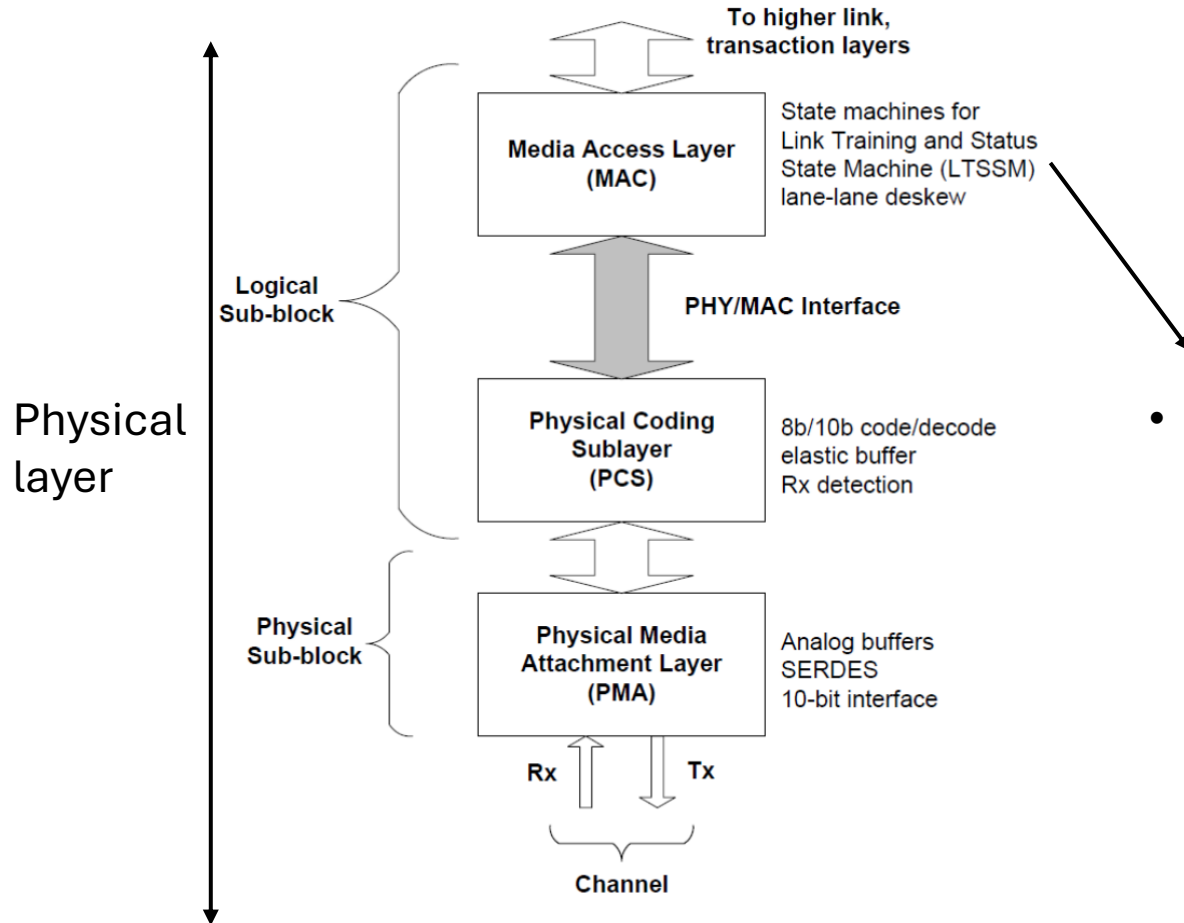


PIPE Overview



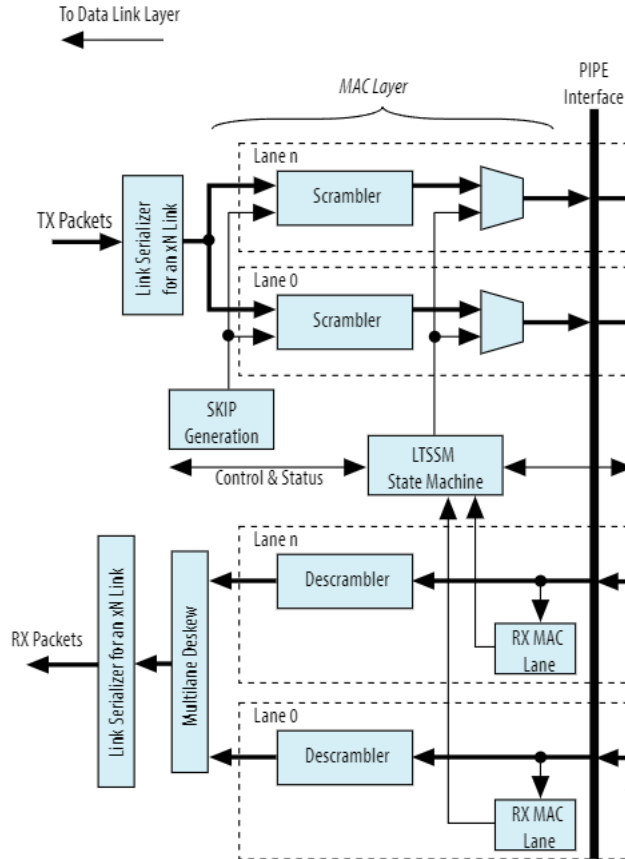
- The **PHY Interface for the PCI Express Architecture (PIPE)** is part of the physical layer and serves as an interface between the Media Access Layer (MAC) and the Physical Coding Sublayer (PCS).

MAC Overview

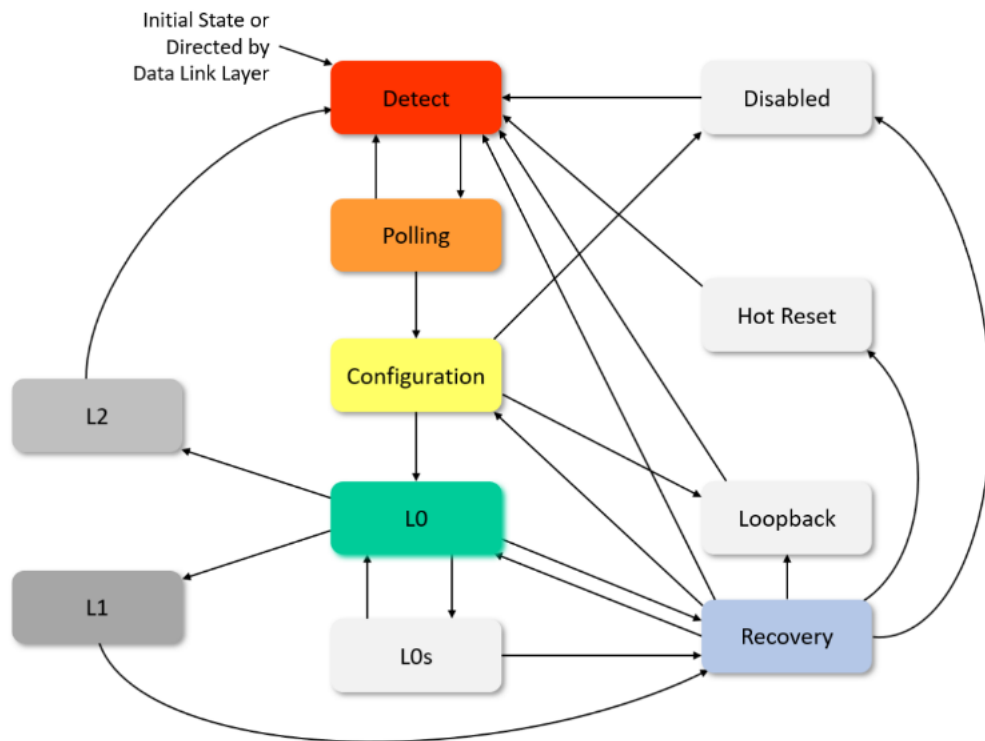


- The **Media Access Layer (MAC)** is part of the physical layer. It lies in between the **PIPE Interface** and the **Data Link Layer (DLL)**.

MAC Overview

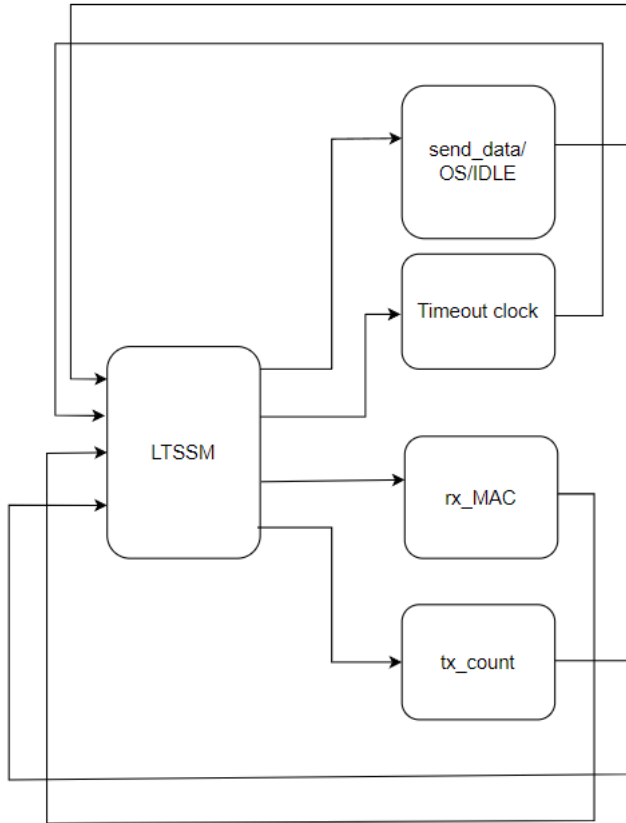


- Scrambler/Descrambler to decode/encode data.
- SKP Generator to insert SKP Ordered Set at an interval between 1180 and 1538 Symbol Times, compensating different bit rates of two ports.
- LTSSM to control Link training and get the Link up.



LTSSM Block Diagram

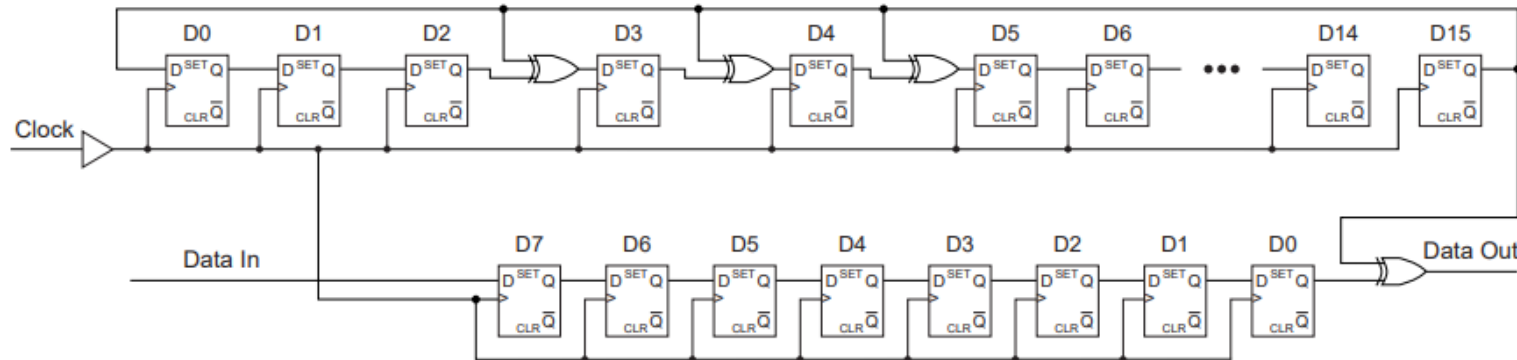
LTSSM Logic



- Send_data/OS/IDLE: Assemble the transmitted data, e.g. TS1 and TS2 Ordered Sets.
- Timeout clock: Trigger timeout clock at each state according to the Specs.
- Rx_MAC:
 - Detect Link/Lane number from received Ordered Sets.
 - Check if received Ordered Sets are valid based on previously determined Link/Lane number.
 - Count the number of consecutive correctly received Ordered Sets and feedback the count flag to the LTSSM.
 - Detect and remove SKP Ordered Sets from received sequence.
- Tx_count: Count the number of transmitted Ordered Set and feedback the count flag to LTSSM

Scrambling/Descrambling

- Scrambling and Descrambling implemented with LFSR.
- Scrambling and Descrambling is done for each symbol.
- LFSR polynomial: $X^{16}+X^5+X^4+X^3+1$
- Rules:
 - COM initializes LFSR.
 - Initialized value is 16'hFFFF.
 - LFSR advances eight shifts for each symbol except for SKP.
 - All data characters except for those in Training Sequence are scrambled.
 - All special characters (K symbols) are not scrambled.
 - Scrambling can only be disabled at the end of Configuration.



Polarity Inversion



- In case of polarity inversion, COM symbol is still correctly detected. Thus, Alignment still works.
- During training sequence, Symbols 6-15 of TS1 and TS2 will be considered to determine if Polarity inversion is necessary or not.
- TS1 Symbols 6-15 received will be D21.5 as opposed to the expected D10.2.
- Symbols 6-15 of the TS2 ordered set will be D26.5 as opposed to the expected D5.2.
- This will be signaled to the RX of SERDES using input port **RX_POLARITY_I** if inversion is needed. All data after that will be non-inverted.
- Transmitter does not have to consider polarity inversion according to the Specs to simplify the implementation.

References

- [PHY Interface for the PCI Express* Architecture \(PIPE\) Specs](#)
- [Simon Southwell's PCIe Primer](#)
- [PCI Express* Solutions User Guide](#)

Contact

Luong Vinh Hai, B.Sc.

Cologne Chip AG

Eintrachtstr. 113

50668 Köln

